THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 6: **Number Play** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** even	Use the even/odd rules.
2	**(b)** 11	Add all the digits.
3	**(c)** 5743	Write the digits in the opposite order.
4	**(d)** 1331	A palindrome reads the same from both ends.
5	**(a)** 1827	A number is divisible by 9 if its digit sum is.
6	**(a)** 22	Check whether the number is even or odd.
7	**(a)** 36	The sum of the first n odd numbers is n^2.
8	**(d)** 74	Value $= 10\times$ tens $+$ units.
9	**(d)** odd	Use the even/odd rules.
10	**(c)** 16	Add all the digits.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Apply the rule to each number in turn; Sequence: $6, 3, 10, 5, 16, 8, 4, 2, 1$ (1); It reaches 1 after 8 steps (1).	1+1
12	Add the number to its reverse; $29 + 92 = 121$ (1); Palindrome obtained: 121 (1).	1+1
13	Digit sum $= 18$; Divisible by 3: yes (digit sum is a multiple of 3) (1); Divisible by 9: yes (digit sum is a multiple of 9) (1).	1+1
14	Apply the rule to each number in turn; Sequence: $5, 16, 8, 4, 2, 1$ (1); It reaches 1 after 5 steps (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Largest $= 5432$ (1); Smallest $= 2345$ (digits arranged in increasing order) (1); Difference $= 5432 - 2345 = 3087$ (1).	1+1+1
16	Middle number $= 48 \div 3 = 16$ (1); The numbers are 15, 16 and 17 (1); Check: $15 + 16 + 17 = 48$ (1).	1+1+1
17	13×7 is odd (odd $\times$ odd) (1); 4 is even (1); The sum is odd (1).	1+1+1
	OR: Largest $= 8531$ (1); Smallest $= 1358$ (digits arranged in increasing order) (1); Difference $= 8531 - 1358 = 7173$ (1).	1+1+1
18	Middle number $= 75 \div 3 = 25$ (1); The numbers are 24, 25 and 26 (1); Check: $24 + 25 + 26 = 75$ (1).	1+1+1
	OR: 12×9 is even (even $\times$ odd) (1); 5 is odd (1); The sum is odd (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Reverse $= 62$ (1); Sum $= 26 + 62 = 88$ (1); $88 = 11 \times 8$ (1); Divisible by 11; quotient $= 8$ (2).	1+1+1+2
	OR: Sequence: $9, 28, 14, 7, 22, 11, 34, 17, 52, 26, 13, 40, 20, 10, 5, 16, 8, 4, 2, 1$ (1); Steps $= 19$ (1); Largest number $= 52$ (1); Even numbers: 13 (1); Ends in the loop $4, 2, 1$ (1).	1+1+1+1+1
20	Reverse $= 74$ (1); Sum $= 47 + 74 = 121$ (1); $121 = 11 \times 11$ (1); Divisible by 11; quotient $= 11$ (2).	1+1+1+2
	OR: Sequence: $11, 34, 17, 52, 26, 13, 40, 20, 10, 5, 16, 8, 4, 2, 1$ (1); Steps $= 14$ (1); Largest number $= 52$ (1); Even numbers: 10 (1); Ends in the loop $4, 2, 1$ (1).	1+1+1+1+1

Mistakes to avoid:

- Saying the sum of two odd numbers is odd; it is even.
- Treating the product of two odd numbers as even; it is odd.
- Writing the reverse of $10a + b$ as $10a + b$ again; the reverse is $10b + a$.
- Using the divisibility test for 9 as if it were for 3 (digit sum 12 is divisible by 3 but not 9).
- In Kaprekar's routine, forgetting to keep leading zeros when the smaller arrangement starts with 0.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 6: **Number Play** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 8021	Write the digits in the opposite order.
2	**(d)** 4554	A palindrome reads the same from both ends.
3	**(c)** 4563	A number is divisible by 9 if its digit sum is.
4	**(a)** 6	Check whether the number is even or odd.
5	**(c)** 81	The sum of the first n odd numbers is n^2.
6	**(a)** 39	Value $= 10\times$ tens $+$ units.
7	**(b)** odd	Use the even/odd rules.
8	**(d)** 17	Add all the digits.
9	**(b)** 1509	Write the digits in the opposite order.
10	**(c)** 27872	A palindrome reads the same from both ends.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Add the number to its reverse; $48 + 84 = 132$; $132 + 231 = 363$ (1); Palindrome obtained: 363 (1).	1+1
12	Digit sum $= 15$; Divisible by 3: yes (digit sum is a multiple of 3) (1); Divisible by 9: no (digit sum is not a multiple of 9) (1).	1+1
13	Apply the rule to each number in turn; Sequence: $10, 5, 16, 8, 4, 2, 1$ (1); It reaches 1 after 6 steps (1).	1+1
14	Add the number to its reverse; $35 + 53 = 88$ (1); Palindrome obtained: 88 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Largest $= 7421$ (1); Smallest $= 1247$ (digits arranged in increasing order) (1); Difference $= 7421 - 1247 = 6174$ (1).	1+1+1
16	Middle number $= 102 \div 3 = 34$ (1); The numbers are 33, 34 and 35 (1); Check: $33 + 34 + 35 = 102$ (1).	1+1+1
17	15×11 is odd (odd $\times$ odd) (1); 8 is even (1); The sum is odd (1).	1+1+1
	OR: Largest $= 9630$ (1); Smallest $= 369$ (digits arranged in increasing order) (1); Difference $= 9630 - 369 = 9261$ (1).	1+1+1
18	Middle number $= 123 \div 3 = 41$ (1); The numbers are 40, 41 and 42 (1); Check: $40 + 41 + 42 = 123$ (1).	1+1+1
	OR: 8×6 is even (even $\times$ even) (1); 9 is odd (1); The sum is odd (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Reverse $= 83$ (1); Sum $= 38 + 83 = 121$ (1); $121 = 11 \times 11$ (1); Divisible by 11; quotient $= 11$ (2).	1+1+1+2
	OR: Sequence: $12, 6, 3, 10, 5, 16, 8, 4, 2, 1$ (1); Steps $= 9$ (1); Largest number $= 16$ (1); Even numbers: 7 (1); Ends in the loop 4, 2, 1 (1).	1+1+1+1+1
20	Reverse $= 95$ (1); Sum $= 59 + 95 = 154$ (1); $154 = 11 \times 14$ (1); Divisible by 11; quotient $= 14$ (2).	1+1+1+2
	OR: Sequence: $13, 40, 20, 10, 5, 16, 8, 4, 2, 1$ (1); Steps $= 9$ (1); Largest number $= 40$ (1); Even numbers: 7 (1); Ends in the loop 4, 2, 1 (1).	1+1+1+1+1

Mistakes to avoid:

- Saying the sum of two odd numbers is odd; it is even.
- Treating the product of two odd numbers as even; it is odd.
- Writing the reverse of $10a + b$ as $10a + b$ again; the reverse is $10b + a$.
- Using the divisibility test for 9 as if it were for 3 (digit sum 12 is divisible by 3 but not 9).
- In Kaprekar's routine, forgetting to keep leading zeros when the smaller arrangement starts with 0.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 6: **Number Play** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 2358	A number is divisible by 9 if its digit sum is.
2	**(b)** 28	Check whether the number is even or odd.
3	**(b)** 144	The sum of the first n odd numbers is n^2.
4	**(b)** 80	Value $= 10\times$ tens $+$ units.
5	**(b)** even	Use the even/odd rules.
6	**(a)** 18	Add all the digits.
7	**(b)** 4865	Write the digits in the opposite order.
8	**(b)** 90609	A palindrome reads the same from both ends.
9	**(c)** 7893	A number is divisible by 9 if its digit sum is.
10	**(d)** 10	Check whether the number is even or odd.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Digit sum $= 19$; Divisible by 3: no (digit sum is not a multiple of 3) (1); Divisible by 9: no (digit sum is not a multiple of 9) (1).	1+1
12	Apply the rule to each number in turn; Sequence: $3, 10, 5, 16, 8, 4, 2, 1$ (1); It reaches 1 after 7 steps (1).	1+1
13	Add the number to its reverse; $67 + 76 = 143$; $143 + 341 = 484$ (1); Palindrome obtained: 484 (1).	1+1
14	Digit sum $= 15$; Divisible by 3: yes (digit sum is a multiple of 3) (1); Divisible by 9: no (digit sum is not a multiple of 9) (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Largest $= 7520$ (1); Smallest $= 257$ (digits arranged in increasing order) (1); Difference $= 7520 - 257 = 7263$ (1).	1+1+1
16	Middle number $= 87 \div 3 = 29$ (1); The numbers are 28, 29 and 30 (1); Check: $28 + 29 + 30 = 87$ (1).	1+1+1
17	21×3 is odd (odd $\times$ odd) (1); 7 is odd (1); The sum is even (1).	1+1+1
	OR: Largest $= 9631$ (1); Smallest $= 1369$ (digits arranged in increasing order) (1); Difference $= 9631 - 1369 = 8262$ (1).	1+1+1
18	Middle number $= 156 \div 3 = 52$ (1); The numbers are 51, 52 and 53 (1); Check: $51 + 52 + 53 = 156$ (1).	1+1+1
	OR: 14×5 is even (even $\times$ odd) (1); 10 is even (1); The sum is even (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Reverse $= 41$ (1); Sum $= 14 + 41 = 55$ (1); $55 = 11 \times 5$ (1); Divisible by 11; quotient $= 5$ (2).	1+1+1+2
	OR: Sequence: $14, 7, 22, 11, 34, 17, 52, 26, 13, 40, 20, 10, 5, 16, 8, 4, 2, 1$ (1); Steps $= 17$ (1); Largest number $= 52$ (1); Even numbers: 12 (1); Ends in the loop 4, 2, 1 (1).	1+1+1+1+1
20	Reverse $= 86$ (1); Sum $= 68 + 86 = 154$ (1); $154 = 11 \times 14$ (1); Divisible by 11; quotient $= 14$ (2).	1+1+1+2
	OR: Sequence: $15, 46, 23, 70, 35, 106, 53, 160, 80, 40, 20, 10, 5, 16, 8, 4, 2, 1$ (1); Steps $= 17$ (1); Largest number $= 160$ (1); Even numbers: 12 (1); Ends in the loop 4, 2, 1 (1).	1+1+1+1+1

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